

Reflection on Cambridge Road Safety Evidence

PS01 | What the data supports and what it cannot prove

This project shows how public data can support a focused decision: where Cambridge should begin pedestrian and cyclist safety investigations. The 2023–2025 CPD records identify 696 eligible crashes and a provisional shortlist of six intersections with at least six reported crashes each. Massachusetts Avenue and Albany Street has the highest count among the explicitly labeled pairs, with nine. Retaining all three locations tied at six avoids an arbitrary cutoff at five candidates.

The most consequential limitation is location quality. Although 430 crashes have two intersection street names, 266 do not. Spelling variants can also split records for the same physical place. The website's current map covers only 322 crashes, making a map marker an incomplete view of the evidence. A location absent from the map cannot be assumed safe. This is why the shortlist should remain provisional while addresses and street names are reconciled.

The data supports comparisons of reported counts and descriptions of recurrence. For example, Massachusetts Avenue and Windsor Street recorded two, three and three eligible crashes across the three years. Columbia Street and Hampshire Street recorded five, one and zero. These patterns help frame audit questions, but they cannot establish whether risk increased or decreased. Changes in walking and cycling volumes, reporting, or street design could explain part of the difference. Zero reported crashes in a year does not establish safety.

The analysis cannot identify the most dangerous location per traveler, prove the cause of a crash, or estimate the benefit of a particular intervention. It lacks exposure counts, verified site conditions and a causal evaluation design. Injury flags concern anyone involved and do not establish severity. The overlapping source files also show why data preparation matters: adding them would count many incidents twice.

The defensible next steps are to audit the shortlisted intersections and improve the location and exposure evidence before allocating capital. Audits should examine current crossing conditions and turning interactions, while checking recent and planned street changes. Publishing aggregate results and clearly explaining uncertainty makes the website useful without overstating what the data proves. Its value is to guide better questions and targeted investigation, with final engineering decisions informed by further evidence.

Basis: the supplied Cambridge crash CSVs, the project methodology, and the local website preview inspected September 19, 2026. The analysis remains descriptive and provisional.